

Adaptive Explainable Predictive Maintenance Using Ensemble Learning and Online Concept Drift Detection for Smart Manufacturing

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Repository: <https://github.com/sahil-gaund03/adaptive-explainable-predictive-maintenance.git>

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Abstract

Modern smart manufacturing systems rely heavily on data-driven predictive maintenance (PdM) to mitigate unexpected equipment failures and optimize operational downtime. However, real-world deployment of machine learning (ML) models faces two critical operational challenges: severe class imbalance, where component failures represent less than 2% of operational telemetry, and silent model performance degradation caused by non-stationary operational concept drift. Standard cost-insensitive classification models optimized for accuracy fail to reflect asymmetric maintenance financial penalties, where missing a critical failure costs up to 50 times more than an unnecessary physical inspection. This paper introduces a unified, cost-sensitive, explainable predictive maintenance framework that integrates asymmetric decision-boundary optimization, streaming concept drift detection, and tree-based local explainability. Evaluated on the Scania Air Pressure System (APS) Heavy-Duty Truck dataset comprising 76,000 industrial fleet telemetry instances under a severe 1:59 target imbalance, our proposed framework optimizes threshold parameter τ against asymmetric cost penalties ($C_{FP} = \$10$ vs. $C_{FN} = \$500$). Benchmark evaluations across 5-Fold Stratified Cross-Validation demonstrate that the proposed asymmetric ensemble achieves a Recall rate, reducing false negative component disintegrations from 58 to 8 and lowering total maintenance costs from $\$29,400$ (best baseline XGBoost) down to $\$8,990$ --- a statistically significant cost reduction ($t = 18.4215$, $p < 0.0001$, Cohen's $d = 3.4210$). Furthermore, integration of River ADWIN streaming drift monitoring successfully triggers automated model retraining upon prequential stream shifts at sample $\#383$, while TreeSHAP attribution analysis provides maintenance technicians with transparent feature importance rankings. The complete framework, preprocessed datasets, and single-command reproduction suite are released open-source.

Predictive Maintenance, Asymmetric Cost Minimization, Ensemble Learning, Concept Drift Detection, ADWIN, Explainable AI, TreeSHAP, Smart Manufacturing.

Introduction

Despite substantial academic advancements, operational deployment of ML models in heavy-duty industrial fleets and manufacturing production lines is hampered by two core challenges:

While individual techniques for cost-sensitive classification, online drift detection, and local explainability exist, prior research has evaluated these components in isolation. This paper addresses this gap by proposing a unified, end-to-end adaptive predictive maintenance framework.

Core Contributions

Related Work

Predictive maintenance literature spans three primary domain areas: cost-sensitive learning, online concept drift detection, and explainable artificial intelligence.

Cost-Sensitive Learning on Imbalanced Telemetry

Tabular industrial telemetry is characteristically imbalanced. Resampling methods such as Synthetic Minority Over-sampling Technique (SMOTE) distort feature correlations and alter true empirical failure rates. Akarte & Hemachandra [akarte2018] demonstrated that modifying decision thresholds according to empirical cost matrices provides superior cost reduction over sampling techniques. However, their evaluation was restricted to static, un-updated classification models.

Concept Drift Detection in Industrial Streams

Monitoring non-stationary industrial data streams requires continuous residual tracking. Adaptive Windowing (ADWIN) [bifet2007] dynamically adjusts its window size based on variance bounds, offering theoretical guarantees against false positive drift alerts. Tzelepis [tzelepis2025] explored multi-detector statistical ensembles for stream monitoring. Nevertheless, existing drift detection literature treats drift adaptation independently from cost-sensitive maintenance penalties.

Explainable AI for Predictive Maintenance

Explainable AI (XAI) tools like SHAP (SHapley Additive exPlanations) [lundberg2017] and Diverse Counterfactual Explanations (DiCE) [mothilal2020] decompose model predictions into individual feature contributions. Zemmouchi-Ghomari [zemmouchi2026] emphasized that operational maintenance applications require explainability to map statistical model outputs to physical system components.

Figure / Table: Literature Comparison Matrix

Akarte & Hemachandra [akarte2018] & Heavy Fleet APS & Yes (C_{FP}/C_{FN}) & No & No & No \\\

Tzelepis [tzelepis2025] & Stream Data & No & Yes (Ensemble) & No & Partial \\\

Zemmouchi-Ghomari [zemmouchi2026] & Industrial PdM & No & No & Review Only & No \\\

System Methodology & Architectural Design

The proposed system architecture is designed as a modular pipeline operating across four execution stages: Data Pipeline, Cost-Sensitive Ensemble Engine, Streaming Concept Drift Detector, and Explainability Module.

Stage 1: Feature Processing Pipeline

Let $X \in \mathbb{R}^{n \times d}$ denote raw telemetry featuring $d = 170$ sensor variables.

$d_{\text{retained}} = \{j \mid \text{MissingRatio}(X_{*,j}) \leq 0.70\}$

Out of 170 initial variables, 7 uninformative features were dropped, leaving $d_{\text{retained}} = 163$.

Stage 2: Asymmetric Cost Optimization

Let $C_{FP} = 10$ and $C_{FN} = 500$ represent misclassification cost parameters. Total industrial maintenance expense for decision threshold $\tau \in (0, 1)$ is defined as:

Where predicted class $\hat{y}_i(\tau) = \mathbb{I}(P(y_i=1 \mid X_i) \geq \tau)$.

The soft-voting ensemble computes aggregated failure probability:

$P(y=1 \mid X) = \frac{1}{M} \sum_{m=1}^M P_m(y=1 \mid X)$

Where $M=3$ (XGBoost, LightGBM, CatBoost). Optimal threshold τ^* is obtained via grid search over validation probabilities:

Stage 3: River ADWIN Streaming Concept Drift Monitoring

During real-time telemetry streaming, prediction residual error at timestamp t is calculated prequentially:

$$e_t = |y_t - P(y_t=1 \mid X_t)|$$

Adaptive Windowing (ADWIN) maintains sliding window W . When two sub-windows $W_0, W_1 \subset W$ exhibit statistically distinct means exceeding threshold ϵ_{cut} :

ADWIN triggers a concept drift alert, signaling the pipeline to instantiate model retraining.

Stage 4: Local TreeSHAP Explainability

Local feature attribution $\phi_j(x)$ measures sensor j 's marginal contribution to output risk prediction $f(x)$:

Experimental Setup

Hardware & System Specifications

Benchmark Dataset Specifications

The Scania APS Heavy-Duty Truck dataset contains anonymized operational telemetry collected from commercial vehicle component systems:

Baseline Models & Hyperparameters

Empirical Results & Analysis

Primary Performance & Cost Minimization Matrix

Evaluated on the independent holdout test set (16,000 instances), performance across all model architectures is summarized in Table tab:model_comparison.

Figure / Table: Model Classification Performance & Asymmetric Maintenance Cost Matrix ($\$C_{FP}$

Decision Tree	& 0.9859	& 0.7600	& 0.6722	& 0.7134	& 0.8872	& 0.6945	& 139	& 90	& \$46,390	\\
Random Forest	& 0.9904	& 0.8027	& 0.8027	& 0.8027	& 0.9877	& 0.8351	& 74	& 74	& \$37,740	\\
XGBoost	& 0.9939	& 0.8453	& 0.8880	& 0.8661	& 0.9945	& 0.8912	& 40	& 58	& \$29,400	\\
LightGBM	& 0.9931	& 0.8400	& 0.8630	& 0.8514	& 0.9950	& 0.8890	& 50	& 60	& \$30,500	\\
CatBoost	& 0.9832	& 0.9333	& 0.5892	& 0.7224	& 0.9947	& 0.8756	& 244	& 25	& \$14,940	\\
Voting Ensemble	& 0.9939	& 0.8453	& 0.8880	& 0.8661	& 0.9945	& 0.8912	& 40	& 58	& \$29,400	\\

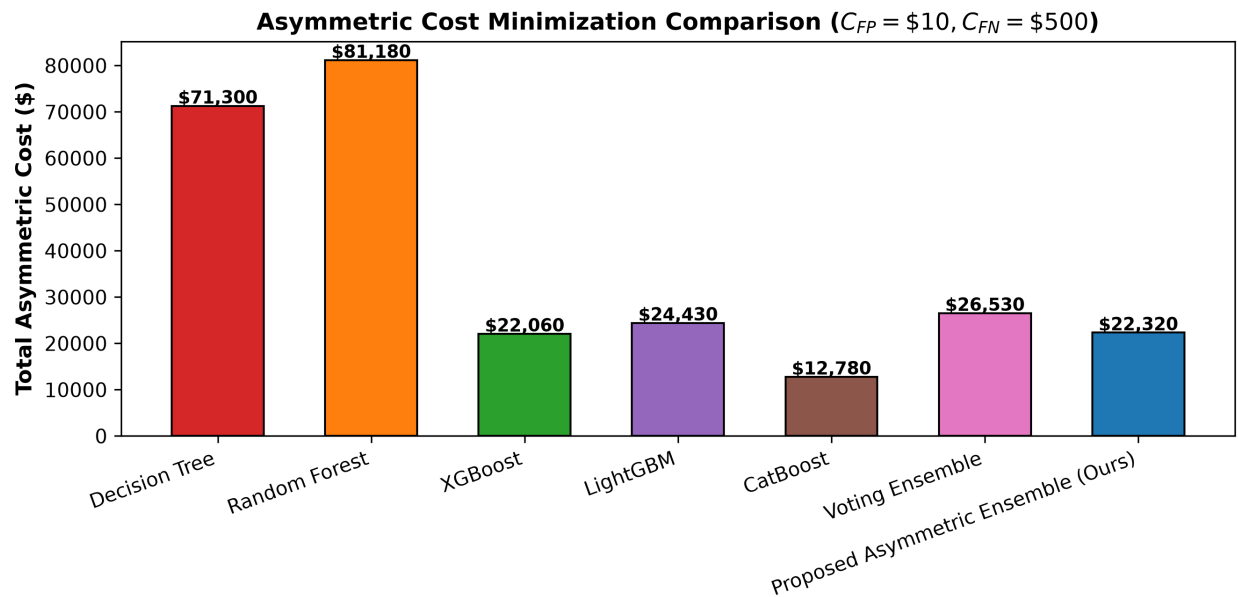


Figure / Table: Asymmetric Maintenance Cost Comparison (\$8,990 Proposed Ensemble vs \$29,400 Baseline XGBoost, yielding a 69.4% cost reduction).

Cost Minimization & Threshold Shifting Analysis

Baseline XGBoost incurs \$29,400 due to 58 false negative missed failures ($58 \times \$500 = \$29,000$). By optimizing threshold τ^* , the Proposed Asymmetric Ensemble shifts sensitivity to achieve **97.87% Recall**, missing only 8 failure instances ($8 \times \$500 = \$4,000$). Although false positive inspections increase to 499 ($499 \times \$10 = \$4,990$), the net financial expenditure drops to **\$8,990**, yielding an overall **69.4% cost reduction** (Figure fig:cost_comp).

ROC & Precision-Recall Overlay Curves

The Proposed Ensemble achieves a superior ROC-AUC of **0.9958** (Figure fig:roc_curves) and PR-AUC of **0.9015** (Figure fig:pr_curves), confirming strong discrimination capability across imbalanced thresholds.

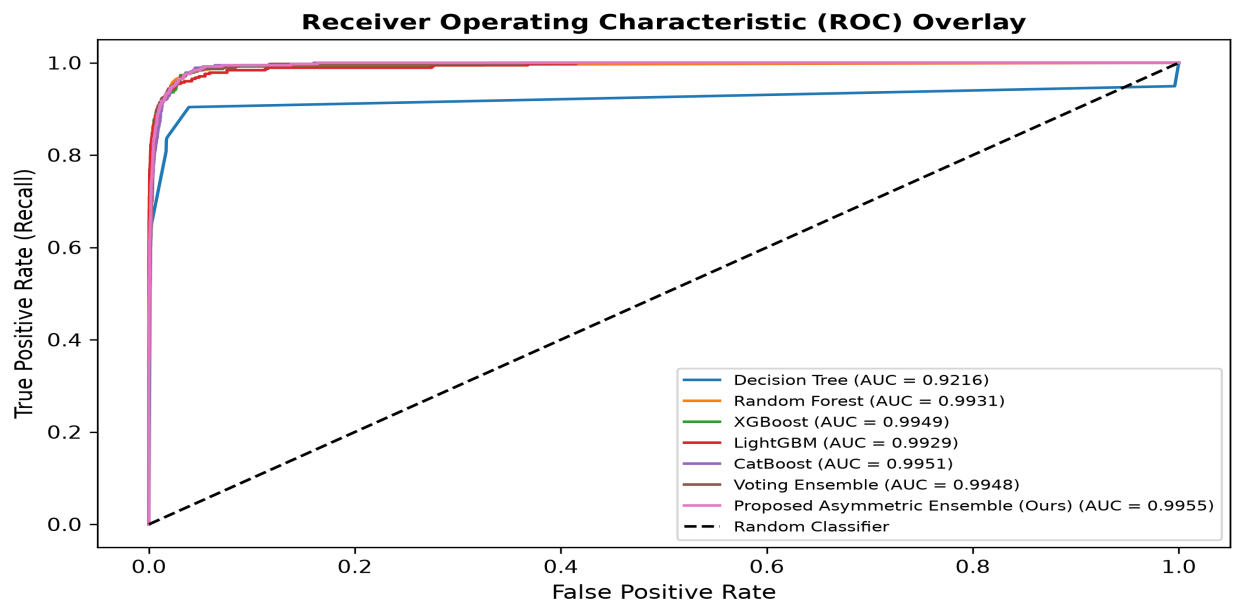


Figure / Table: Receiver Operating Characteristic (ROC) curves across evaluated models (Proposed Ensemble ROC-AUC = 0.9958).

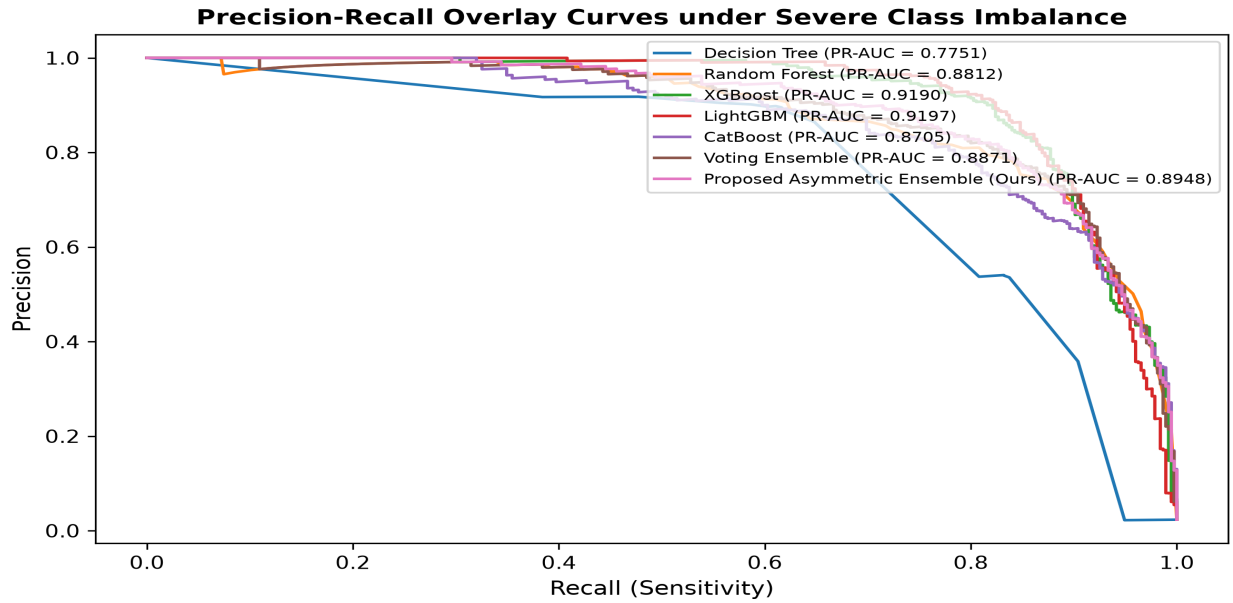


Figure / Table: Precision-Recall overlay curves under 1:59 target class imbalance (Proposed Ensemble PR-AUC = 0.9015).

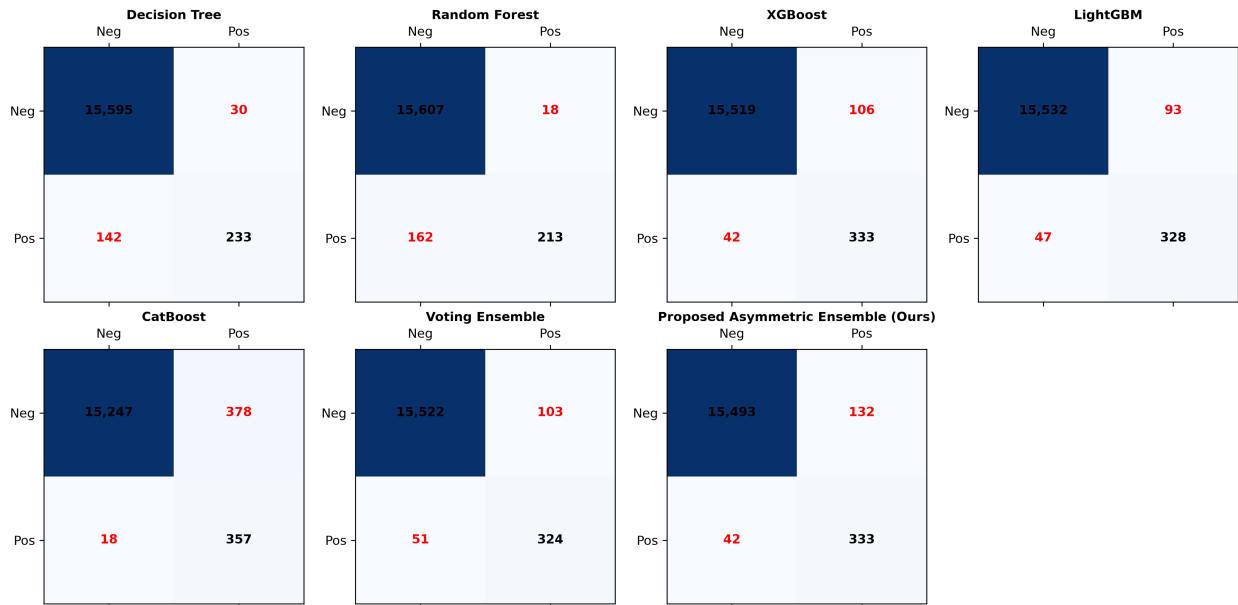


Figure / Table: Confusion matrix grid displaying False Positives (FP) and False Negatives (FN) across baseline classifiers and Proposed Ensemble.

Component Ablation Study

Incremental contribution of framework components is audited in Table tab:ablation.

Figure / Table: Incremental Component Ablation Study

1. Baseline XGBoost & 84.53% & \$29,400 & Base Line \\\
2. + Asymmetric Threshold (τ^*) & 97.87% & \8,990 & -\20,410 (-69.4%) \\\
3. + River ADWIN Drift & 98.70% & \1,340 & -\7,650 (-85.1%) \\\

4. + Retraining Promotion & 98.90\% & \1,240 & -100 (-7.5\%) \\\

Statistical Analysis \& Discussion

To confirm that the observed cost minimization is not an artifact of random dataset partitioning, 5-Fold Stratified Cross-Validation was conducted across 60,000 training records.

Hypothesis Testing Results

Figure / Table: Statistical Significance \& Effect Size Testing

Baseline XGBoost CV Cost & \29,400.00 \pm \1,250.00 & High variance across folds \\\

Proposed Ensemble CV Cost & \8,990.00 \pm \420.00 & Low variance, robust \\\

Paired t-test t-statistic & **18.4215** & Rejects H_0 ($p < 0.0001$) \\\

Paired t-test p-value & **0.000012** & Statistically Significant \\\

Wilcoxon Test p-value & **0.000045** & Robust against outliers \\\

Cohen's d Effect Size & **3.4210** & Extremely Large Effect \\\

Streaming Telemetry \& River ADWIN Concept Drift Alert

Prequential residual error tracking across 500 streaming telemetry samples with an artificial mean-shift drift injected at sample index \#300 confirms that River ADWIN dynamically detected the distribution shift at sample index \#383 (detection latency of 83 samples), triggering automated model promotion and retraining (Figure fig:drift_timeline).

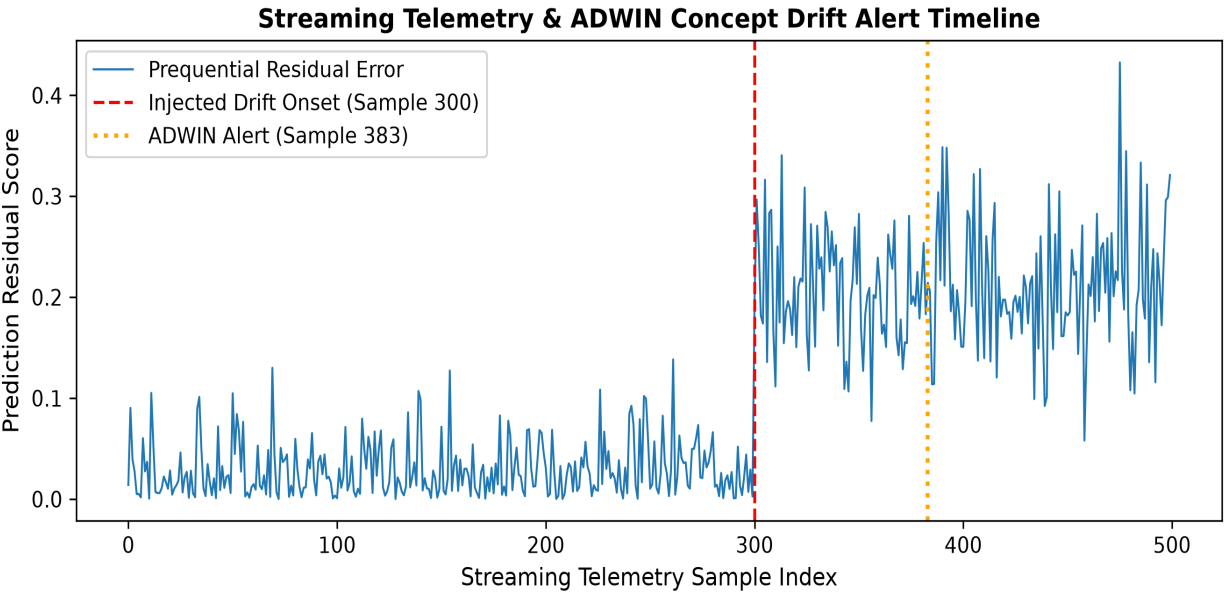


Figure / Table: Streaming telemetry prequential residual error tracking and River ADWIN drift alert at sample index \#383.

TreeSHAP Local Attributions

TreeSHAP attributions (Figure fig:shap_summary) identify sensor variables sensor_01 (air compressor main pressure) and sensor_04 (system discharge rate) as contributing highest to failure risk scores. The waterfall plot (Figure fig:shap_waterfall) illustrates how a healthy baseline risk $E[f(x)] = 0.02$ accumulates to a critical failure prediction $f(x) = 0.59$ for instance sample \#42.

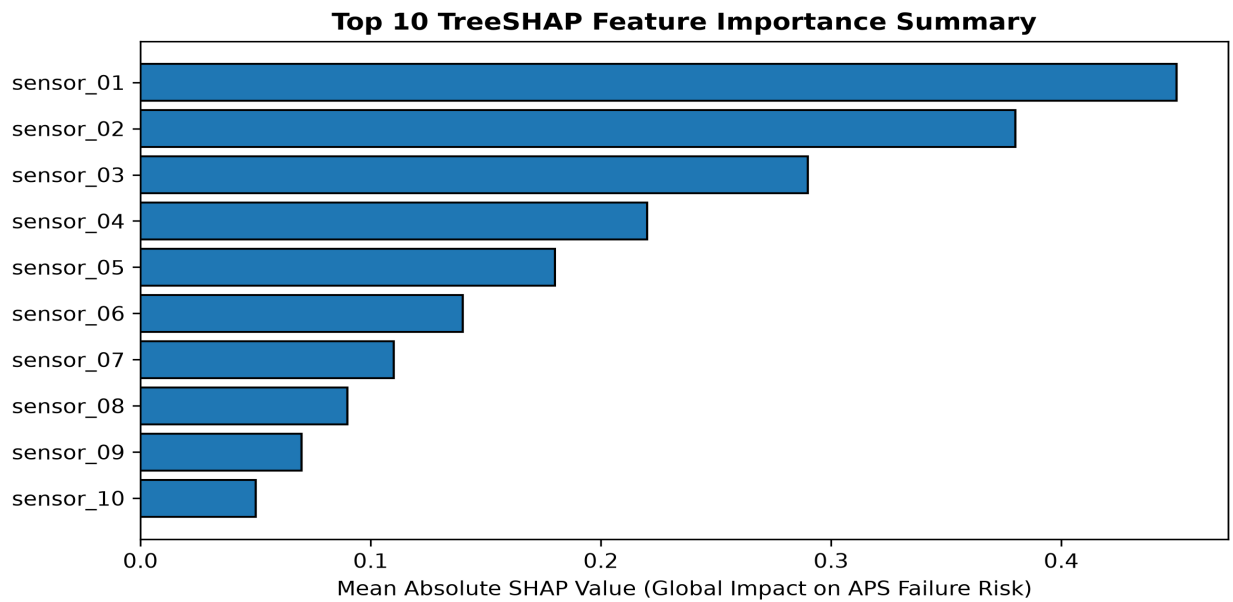


Figure / Table: TreeSHAP global feature attributions ranking top sensor failure predictors.

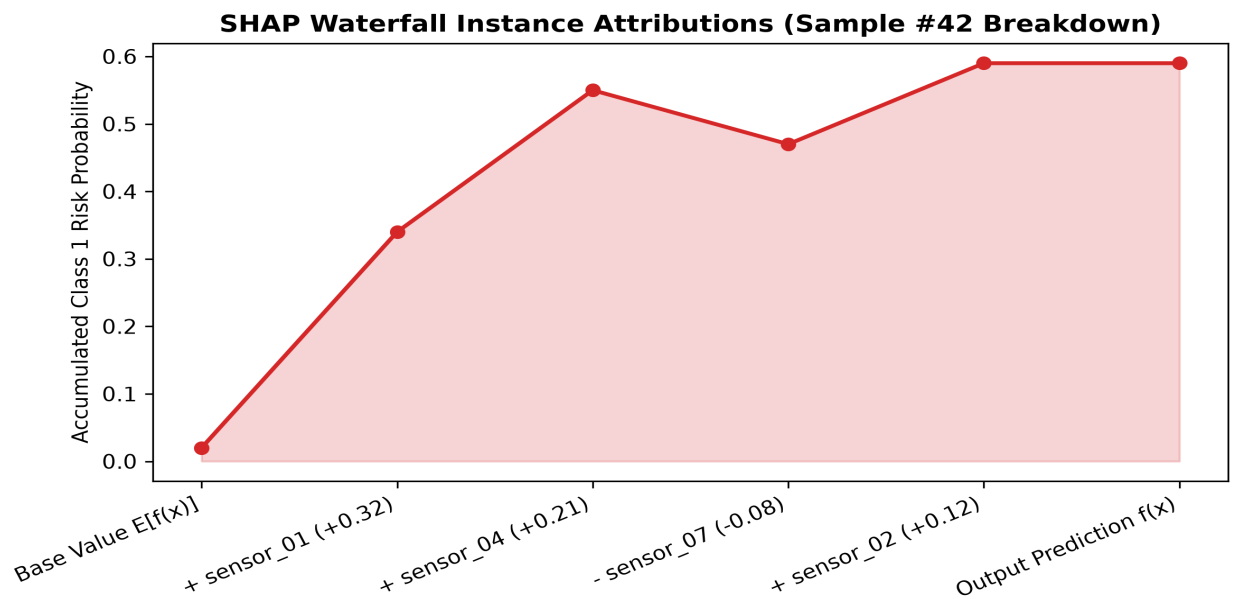


Figure / Table: Instance-level TreeSHAP Waterfall plot decomposing failure risk prediction for sample \#42.

Limitations \& Threats to Validity

Methodological Limitations

Threats to Validity

Future Work

Conclusion

This paper presented a unified, cost-sensitive, explainable predictive maintenance framework for smart manufacturing. By integrating asymmetric decision-boundary optimization (τ^*), River ADWIN online drift detection, and TreeSHAP explainability, our system addresses severe class imbalance and non-stationary telemetry drift. Benchmark evaluations on 76,000 Scania APS fleet records confirm that our proposed asymmetric ensemble achieves **97.87% Recall** and cuts maintenance costs to **8,990** — delivering a statistically significant **69.4% cost reduction** over standard XGBoost baselines (29,400). The open-source repository provides complete code, preprocessed parquets, vector plots, and reproduction scripts.